Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/15/2019
PI Name:	Hoffman, Larry Ph.D.		
Project Title:	Gravity-Induced Plasticity in Mammalian Utricular Hair Cells: Intrinsic or Multisensory?		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
PI Email:	lfh@ucla.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-9823
Organization Name:	University of California Los Angeles		
PI Address 1:	Geffen School of Medicine at UCLA		
PI Address 2:	Box 951624		
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1624	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	03/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 3/31/2020 per F. Hernandez/ARC (Ed., 5/15/19) NOTE: Extended to 3/31/2019 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: Extended to 8/31/2018 per F. Hernandez/ARC (Ed., 7/13/17) NOTE: Extended to 7/31/2017, per NSSC information (Ed., 6/15/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schweizer, Felix Ph.D. (University of California, Los Angeles) Fuller, Charles Ph.D. (University of California, Davis)		
Grant/Contract No.:	NNX13AL99G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal is being submitted to the research topic area Ground based research designed to lead to space flight announced in NASA NRA (NASA Research Announcement) NNH12ZTT001N. We are proposing a series of studies to investigate whether the plasticity in the inner ear utricle is affected by non-vestibular load-sensing physiologic systems. Potential interactions between load-sensing systems might offer cross-modal synergy under altered gravity conditions, e.g., loading of the utricle might directly affect other systems (e.g., immune, osteogenic) that are currently subject to significant countermeasure efforts. In addition, the proposed research endeavors to elucidate how this plasticity is manifested in alterations of the activity of neurons that signal such changes to the central nervous system. We plan to use mouse models that systematically exclude the interaction of particular components of the vestibular load-sensing system. These models represent valuable adjuncts that can be implemented in future flight studies to address countermeasures driven by the recruitment of multiple sensory systems. Furthermore, the present investigation serves as a segue between altered gravity exposure durations that have been focused upon the duration of space shuttle missions and the much longer exposures incumbent with experiments aboard the International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research explores the natural capabilities of utricular hair cells to undergo modifications in their synaptic connections with neurons that communicate sensory signals to the central nervous system as a result of exposure to altered gravitational fields. The ambient gravitational field is a fundamental component of the natural sensory environment, hence a principal aspect of our research is to explore the "sensory learning" that may result when changes in gravity are imposed. These fundamental findings are the foundation for future experiments to reveal the cellular mechanisms responsible for these changes. Since synaptic decline may reflect one form of natural vestibular dysfunction due to aging, pathology, or ototoxicity, an understanding of this repertoire and the cellular mechanisms is the first step toward designing rehabilitative strategies. Our research is designed to develop a systematic laboratory protocol in which these natural capabilities can be explored.
Task Progress:	The principal research achievements made during this funding period include: 1) training of a new postdoctoral scholar recruited to the project and the field of peripheral vestibular neurobiology; 2) development and validation of semi-automated synapse quantification; 3) application of super-resolution microscopy methods for quantification of synapse size, as a proxy for presynaptic architectural heterogeneity; 4) discovery of postsynaptic changes in vestibular afferent neurons in the otoferlin-null mouse (Otof-/-), an important model for vestibular hypofunction which has important implications for exposure to microgravity. The research plans for the coming year include: 1. Complete hypergravity component of the Specific Aims, which include exposing cohorts of WT and Otof-/- to periods of hypergravity, including all analyses. 2. Conduct new analyses of fine dendritic structure within the striolae of WT and Otof-/-. 3. Complete analyses and publication regarding synapse distributions in hair cells from WT and Otof-/- mice. ED. NOTE (5/15/19): Compiled from PI's annual technical report submitted to NASA Ames Research Center in March 2019; covers period from August 2017-March 2019.
Bibliography Type:	Description: (Last Updated: 10/05/2022)
Articles in Peer-reviewed Journals	Prins TJ, Saldate JJ, Berke GS, Hoffman LF. "On the legacy of genetically-altered mouse models to explore vestibular function: Distribution of vestibular hair cell phenotypes in the otoferlin-null mouse." Ann Otol Rhinol Laryngol. 2019 Jun;128(6 Suppl):125S-133S. https://doi.org/10.1177/0003489419834596 ; PMID: 31092028 [note reported previously in May 2019 as "in press as of May 2019"] , Jun-2019